

Saurav Bhattarai

Hydrologist / Physically Based Modeling and
Earth Observation

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Research Interests

Physically based hydrological modeling and flood routing; Earth observation of the water balance (precipitation, soil moisture, snow, and water storage); downscaling and bias correction of satellite products; compound hydroclimatic extremes; open research software.

Education

- 2023–2027 **Ph.D. in Engineering**, Jackson State University, Jackson, MS, USA
Department of Civil and Environmental Engineering. Dissertation defense: February 10, 2027; degree expected May 2027.
Dissertation: *Accessing, Refining, and Combining Open Earth Data for a Modular, Parsimonious, Physics-Based Hydrological Model*.
Advisor: Prof. Rocky Talchabhadel. Committee: Dr. Nawa Raj Pradhan (ERDC), Dr. Sadik Khan, Dr. Kejun Wen, Dr. Robert W. Whalin.
- 2023–2025 **M.S. in Engineering (earned en route to the Ph.D.)**, Jackson State University, Jackson, MS, USA, GPA 4.0/4.0
Major: Civil Engineering. Thesis: *Heatwave Dynamics and Urban Cooling: From Risk Assessment to Nature-Based Solutions*.
Advisor: Prof. Rocky Talchabhadel.
- 2017–2022 **Bachelor of Civil Engineering**, Tribhuvan University, Institute of Engineering, Pulchowk Campus, Lalitpur, Nepal, Score 80.14%
- Licensure Fundamentals of Engineering (FE) Civil exam, passed April 2026 (NCEES).

Research Experience

- Aug 2023–Present **Graduate Research Assistant**, Jackson State University, Jackson, MS, USA, Advisor: Prof. Rocky Talchabhadel
- Apply MRRpy without calibration to observed floods of the Bagmati River (Kathmandu Valley, Nepal) and the Blue River (Oklahoma), diagnosing model error through process representation; for example, corrected soil hydraulic conductivity with a published global relationship instead of fitting it to the basin. Currently extending MRRpy with baseflow and groundwater recharge.
 - Model the August 2026 Bhote Koshi–Trishuli outburst flood in the Nepal Himalaya as an upstream boundary inflow (preliminary).
 - Detected an abrupt 2017 shift in IMERG V07 rainfall over the Kathmandu Valley with long-record rain gauges, attributed it to lost gauge input, and developed a correction for extreme-rainfall estimation.
 - Downscaled SMAP soil moisture from 9 to 1 km and evaluated it inside the satellite pixel against airborne campaigns and storm responses.
 - Mapped the timing of the wettest days over global land with ERA5 (1980–2024) [J13], built a global record of compound heatwave–drought (1950–2025), and analyzed compound flooding from storm surge and rainfall at 79 U.S. tide gauges (1950–2025).
 - Led development of the GeoClimate Intelligence Platform [J9, P1]; completed M.S. research on heat risk and urban cooling [J4, J6, J8, J11]; evaluated satellite precipitation products across the CONUS and Hawaii [J3].
- Oct 2024–Aug 2026 **ORISE Research Fellow**, U.S. Army Engineer Research and Development Center, Coastal and Hydraulics Laboratory, Vicksburg, MS, USA, Mentor: Dr. Nawa Raj Pradhan
- Developed MRRpy, an open-source, distributed, physically based rainfall–runoff and flood-routing model with interchangeable runoff generation (Green–Ampt, variable source area, SCS curve number, impervious surfaces) and routing (kinematic, Muskingum–Cunge, semi-implicit diffusive), Google Earth Engine data retrieval, and CPU/GPU backends.
 - Built and tested a GSSHA model of the Kathmandu Valley (Bagmati River basin) as an established physically based reference.
 - Implemented the ERDC GLOBAL-SAM snow-depth and SERVES root-zone soil-moisture algorithms as Google Earth Engine applications, and evaluated SERVES against SMAP and 2,643 in situ stations worldwide (ISMN, SCAN, SNOTEL, USCRN, NEON).
 - Developed an explainable tree–neural network ensemble that downscales GRACE terrestrial water storage from 55 to 5 km over the Mississippi River Basin, validated against USGS aquifer storage change and 825 groundwater wells.

Publications

Name in bold. First-author journal articles: [J3], [J4], [J6], [J8], [J9], [J11], [J13]. Google Scholar (September 2026): 130 citations, h-index 7.

Peer-Reviewed Journal Articles

- [J14] Gautam, A., **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). Urban morphology controls thermal and moisture anomalies: Insights from local climate zones across U.S. cities. *Remote Sensing Applications: Society and Environment*, 43, 102161. doi:10.1016/j.rsase.2026.102161
- [J13] **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). Do the wettest days occur together? A global analysis on disentangling precipitation intensity from seasonal timing. *Earth and Space Science*, 13(4), e2025EA004845. doi:10.1029/2025EA004845
- [J12] Poudel, S., Bista, S., **Bhattarai, S.**, Sharma, S., & Talchabhadel, R. (2026). Understanding multi-hazard risk for U.S. coastal cities. *International Journal of Disaster Risk Reduction*, 134, 106023. doi:10.1016/j.ijdrr.2026.106023
- [J11] **Bhattarai, S.**, He, C., Sharma, S., Kumar, S., & Talchabhadel, R. (2026). Rainfall relief or humidity havoc? The boon and curse of precipitation during heatwaves. *Natural Hazards*, 122(5), 172. doi:10.1007/s11069-025-07742-x
- [J10] Bista, S., **Bhattarai, S.**, Sharma, S., & Talchabhadel, R. (2026). Local-scale flood hazard projections in historically vulnerable communities. *Modeling Earth Systems and Environment*, 12(1), 5. doi:10.1007/s40808-025-02599-2
- [J9] **Bhattarai, S.**, Pradhan, N. R., & Talchabhadel, R. (2026). GeoClimate Intelligence Platform: A web-based framework for environmental data analysis. *Environmental Modelling & Software*, 197, 106826. doi:10.1016/j.envsoft.2025.106826
- [J8] **Bhattarai, S.**, Banjara, P., Pandey, V. P., Aryal, A., Pradhan, P., Al-Douri, F., Pradhan, N. R., & Talchabhadel, R. (2025). Quantifying the cooling effects of blue-green spaces across urban landscapes: A case study of Kathmandu Valley, Nepal. *Urban Climate*, 61, 102493. doi:10.1016/j.uclim.2025.102493
- [J7] Talchabhadel, R., Panthi, J., Pandey, V. P., Rakhal, B., Ghimire, G. R., Bista, S., **Bhattarai, S.**, Poudel, S., Bhattarai, Y., Prajapati, R., Thapa, B. R., Nepal, B., & Sharma, S. (2025). Yesterday's extremes, today's new normal: Flood risk in the Kathmandu Valley, Nepal. *Natural Hazards*, 121(16), 19409–19423. doi:10.1007/s11069-025-07524-5
- [J6] **Bhattarai, S.**, Bokati, L., Sharma, S., & Talchabhadel, R. (2025). Understanding spatiotemporal variation of heatwave projections across US cities. *Scientific Reports*, 15, 10643. doi:10.1038/s41598-025-95097-5
- [J5] Talchabhadel, R., **Bhattarai, S.**, & Bista, S. (2025). Projected changes in precipitation extremes across the Mississippi River Basin using the NASA global daily downscaled datasets NEX-GDDP-CMIP6. *International Journal of Climatology*, 45(5), e8748. doi:10.1002/joc.8748
- [J4] **Bhattarai, S.**, Bista, S., Sharma, S., White, L. D., Amini, F., & Talchabhadel, R. (2024). Spatiotemporal characterization of heatwave exposure across historically vulnerable communities. *Scientific Reports*, 14, 20882. doi:10.1038/s41598-024-71704-9
- [J3] **Bhattarai, S.**, & Talchabhadel, R. (2024). Comparative analysis of satellite-based precipitation data across the CONUS and Hawaii: Identifying optimal satellite performance. *Remote Sensing*, 16(16), 3058. doi:10.3390/rs16163058
- [J2] Banjara, P., **Bhattarai, S.**, Pandey, V. P., & Talchabhadel, R. (2024). Spatiotemporal characterization of heatwaves on an urban center using satellite-based estimates. *Theoretical and Applied Climatology*, 155(7), 6677–6695. doi:10.1007/s00704-024-05026-1
- [J1] Bhattarai, T. N., Ghimire, S., Aryal, S., Baaniya, Y., **Bhattarai, S.**, Sharma, S., Bhattarai, P. K., & Pandey, V. P. (2023). Projected changes in hydro-climatic extremes with CMIP6 climate model outputs: A case of rain-fed river systems in Western Nepal. *Stochastic Environmental Research and Risk Assessment*, 37(3), 965–987. doi:10.1007/s00477-022-02312-0

Peer-Reviewed Conference Proceedings

- [P1] **Bhattarai, S.**, Pradhan, N. R., Walters, W. L., & Talchabhadel, R. (2026). Bridging the gap: An open-source, low-code platform for integrating data science into water resources engineering education. In *World Environmental and Water Resources Congress 2026* (pp. 80–91). ASCE. doi:10.1061/9780784486931.007

Manuscripts Under Review and in Preparation

- Under review Kalonji, A., Khalil, S., **Bhattarai, S.**, Talchabhadel, R., & Sharma, S. Multiscale evaluation of IMERG satellite precipitation estimates across the United States.

- In preparation **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. MRRpy: A modular Python model for distributed rainfall–runoff generation and flood routing. To be submitted to *Journal of Open Source Software*.
- In preparation **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Explainable tree–neural network ensemble for GRACE-based terrestrial water storage and groundwater estimation.
- In preparation **Bhattacharai, S.**, Talchabhadel, R., & Pradhan, N. R. An abrupt 2017 shift in IMERG V07 rainfall over the Kathmandu Valley: Detection with long-record gauges, attribution to lost gauge input, and consequences for bias correction and extreme-rainfall estimation.
- In preparation **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. A 76-year global record reveals accelerating compound heatwave–drought hazards in drylands.
- In preparation **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Sea-level rise, not changing storm dependence, explains the rise in compound flood days along the contiguous U.S. coast. Target: *Earth's Future*.

Dataset

- [D1] Talchabhadel, R., & **Bhattacharai, S.** (2024). *Interconnected hazards data* [Data set]. DesignSafe-CL. doi:10.17603/ds2-a2vx-sx21

Research Software

- MRRpy Lead developer. Open-source (MIT) distributed rainfall–runoff and flood-routing model with Python, command-line, and QGIS interfaces, Google Earth Engine data retrieval, and Numba/CuPy (GPU) acceleration; on PyPI (`pip install mrrpy`), with automated tests (GitHub Actions) and documentation including an interactive hydrology course. mrrpy.readthedocs.io; github.com/SauravBhattacharai19/MRRpy
- GeoClimate Lead developer. GeoClimate Intelligence Platform: no-code Google Earth Engine web platform with nine tools (including GLOBAL-SAM snow depth and SERVES soil moisture) and more than 1,000 datasets; used in university courses and faculty workshops [J9, P1]. geoclimate.streamlit.app; github.com/SauravBhattacharai19/GeoClimate-Fetcher
- Earth Engine app Cooling Efficiency Analyzer: estimates the cooling potential of urban green and blue spaces from Landsat land surface temperature [J8]. sauravbhattacharai1999.users.earthengine.app/view/cooling-efficiency-analyzer
- Course platform Advanced Water Resources Engineering: interactive Streamlit course materials. advancedwater.streamlit.app

Fellowships and Grants

- 2026 **NASA FINESST** (Future Investigators in NASA Earth and Space Science and Technology), submitted, pending. “Satellite-Derived Antecedent Soil Moisture Deficit for Calibration-Free Initialization of a Gridded, Physics-Based Flood Model.” Future Investigator; conceived the project and wrote the proposal (\$50,000 requested).
- 2024–2026 **ORISE Research Fellowship**, U.S. Department of Defense Research Participation Program, ERDC Coastal and Hydraulics Laboratory; extended based on research performance.
- 2024–2025 **Co-Investigator**, Microsoft–AUC Mini-Grants Program: “Climate Change Impact Assessment Training for Undergraduate Students: Leveraging Planetary-Scale Data and Cloud.”
- 2023–Present **Graduate Research Assistantship and full scholarship**, Hydrological Impacts Computing, Outreach, and Resiliency Partnership (HICORPS), Jackson State University, with ERDC and Woolpert.

Honors and Awards

- 2026 Academic Excellence Award, International Students Association, Jackson State University.
- 2026 National finalist (team leader), Mastercard IGS × AUC Data Science Initiative Data Challenge (top 26 of 106 teams from 13 HBCUs).
- 2025 Best in Show and first position (lead developer, GeoClimate Intelligence Platform), Microsoft × AUC Data Science Initiative Data Challenge.
- 2025 Honorable Mention (team leader), Mastercard × AUC Data Science Initiative Data Challenge (top 18 of 70+ teams; \$1,000).
- 2024 Winner, AGU Catchment Hydrology Subcommittee Early Career Competition (LSTM rainfall–runoff modeling; \$300).
- 2017–2021 Merit-based full scholarship, Institute of Engineering, Pulchowk Campus (ranked 19th of about 15,000 applicants).

Training and Competitive Programs

- 2026 National Data Platform Education Hub Content Workshop, San Diego Supercomputer Center (selected participant; \$2,000 stipend).
- 2025 CUAHSI-SINTER Snow Measurement Field School, Granby, CO (selected participant; \$600 travel award).
- 2025 NCAR Graduate Visitor Program, Boulder, CO (research residency with NCAR scientists).
- 2025 Tethys Summit, Tampa, FL (travel grant).
- 2024 AGU Chapman Conference on Remote Sensing of the Water Cycle, Honolulu, HI (travel grant).
- 2024 Natural Hazards Center CyberTraining Workshop (travel support).

Teaching Experience

- Fall 2025 **Teaching Assistant**, *Advanced Water Resources Engineering (CIV 465)*, Jackson State University
Designed modules on big-data analytics, machine learning, extreme-value analysis, and climate projections; built the course's interactive platform; delivered lectures and Python/GIS training.
- Spring 2024, 2025 **Teaching Assistant**, *Water Resources Engineering (CIV 370)*, Jackson State University
Taught frequency analysis, unit hydrographs, hydrologic routing, open-channel hydraulics, and stormwater design; designed assignments and exams.
- Spring 2024 **Teaching Assistant**, *Fluid Mechanics Laboratory (CIVL 330)*, Jackson State University
Ran weekly laboratory sessions and graded reports.
- 2022–2023 **Mathematics and Physics Teacher**, *Albright Academy*, Pokhara, Nepal
Variable schedule (about four months in total).

Mentoring

- Howard University (2 undergraduate researchers): evaluation of IMERG precipitation against stations and MRMS; manuscript under review.
- Tribhuvan University, Nepal (5 undergraduate researchers): theses on heatwaves in Nepal.
- Jackson State University (2 undergraduate researchers): development and testing of the GeoClimate Intelligence Platform.
- 2026 Research Mentor, AUC Data Science Initiative PreFreshman Experience (incoming HBCU students; team data-science projects).

Presentations

Oral Presentations (Presenting Author)

- 2026 **Bhattacharai, S.**, Pradhan, N. R., Walters, W. L., & Talchabhadel, R. Bridging the gap: An open-source, low-code platform for integrating data science into water resources engineering education. ASCE-EWRI World Environmental and Water Resources Congress, Mobile, AL, April.
- 2026 **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Democratizing hydroclimatic science education through the GeoClimate Intelligence Platform. 106th AMS Annual Meeting, January.
- 2025 Talchabhadel, R., **Bhattacharai, S.**, Bista, S., Poudel, S., Jones, D., & Terrett, S. Democratizing Earth science data for community-based water hazard assessment: A co-production platform for hydrologic intelligence. AGU Fall Meeting, New Orleans, LA, December.
- 2025 **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Advancing groundwater monitoring and downscaling using open datasets and machine learning in the Mississippi River Basin. XIX World Water Congress, Marrakech, Morocco, December.
- 2025 **Bhattacharai, S.**, & Talchabhadel, R. Evaluating the compound effects of precipitation and storm surge on coastal flooding risk in the CONUS. ASCE-EWRI World Environmental and Water Resources Congress, Anchorage, AK, May.
- 2024 **Bhattacharai, S.**, & Talchabhadel, R. Assessing cascading and compound climate extremes and associated vulnerabilities across the United States. 3rd International Conference on Natural Hazards and Risks in a Changing World, Amsterdam, Netherlands, June.
- 2024 **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Assessing multidimensional climate extremes and associated vulnerabilities across the United States. EGU General Assembly, Vienna, Austria, April (EGU24-20600).
- 2023 **Bhattacharai, S.**, Bokati, L., Sharma, S., & Talchabhadel, R. Assessing heatwave impacts and enhancing climate resilience in highly populated US cities. AGU Fall Meeting, San Francisco, CA, December.

Poster Presentations (Presenting Author)

- 2026 **Bhattacharai, S.**, Pradhan, N. R., & Talchabhadel, R. Explainable tree–neural network ensemble for GRACE-based TWS downscaling and groundwater estimation. ERDC RD26 Workshop, Vicksburg, MS.
- 2025 Talchabhadel, R., **Bhattacharai, S.**, Prajapati, A. K., Karn, B., Bhattacharai, B., Yadav, R., & Bhattacharai, P. K. Spatiotemporal assessment of urban heat island and heatwave dynamics in densely populated Nepalese cities: Evaluating nature-based cooling strategies. AGU Fall Meeting, New Orleans, LA, December.
- 2025 **Bhattacharai, S.**, Talchabhadel, R., & Pradhan, N. R. Precipitation temporal disaggregation for event-based hydrological modeling in the Kathmandu watershed, Nepal. AGU Fall Meeting, New Orleans, LA, December.
- 2024 **Bhattacharai, S.**, Byrd, A., & Talchabhadel, R. Evaluating the compound effects of precipitation and storm surge on coastal flooding risk in the CONUS: A detailed case study of Galveston. AGU Fall Meeting, Washington, DC, December.
- 2024 **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Detecting multihazards in the United States using machine learning: A comparative analysis of individual and compound hazard events. AGU Fall Meeting, Washington, DC, December.
- 2024 Talchabhadel, R., **Bhattacharai, S.**, Banjara, P., & Pandey, V. P. Urban cooling strategies: Evaluating the effectiveness of green spaces and water bodies in mitigating heat exposure in Kathmandu Valley, Nepal (GC51H-0059). AGU Fall Meeting, Washington, DC, December.
- 2024 **Bhattacharai, S.**, & Talchabhadel, R. Heatwaves, humidity, and the water cycle: A remote sensing perspective on climate transformation. AGU Chapman Conference on Remote Sensing of the Water Cycle, Honolulu, HI, February.

Co-Authored Abstracts (Presented by Co-Authors)

- 2025 Pradhan, N. R., **Bhattacharai, S.**, Kloss, G., et al. Global Snow Attribute Mapping Tool (Global-SAM). AGU Fall Meeting, New Orleans, LA. [Oral]
- 2025 Gautam, A., **Bhattacharai, S.**, & Talchabhadel, R. Unraveling urban heat and moisture anomalies across the continental U.S. using local climate zones and Earth observation data. AGU Fall Meeting, New Orleans, LA. [Poster]
- 2025 Dahal, B., **Bhattacharai, S.**, Bhattacharai, P. K., & Talchabhadel, R. Evaluating the applicability of SWOT and SWORD for Nepal's water monitoring: Building a scalable national database. AGU Fall Meeting, New Orleans, LA. [Poster]
- 2025 Talchabhadel, R., **Bhattacharai, S.**, Bista, S., Poudel, S., & Kafoury, R. Projecting extreme temperatures and precipitation for the Gulf Coast region of the United States. ASCE-EWRI World Environmental and Water Resources Congress, Anchorage, AK. [Oral]
- 2024 Talchabhadel, R., Bernard, D., Johnson, K., **Bhattacharai, S.**, Bista, S., Poudel, S., & Kafoury, R. Evaluation of climate change projections using open-source software infrastructure across the Gulf Coast of the United States (NH13B-2293). AGU Fall Meeting, Washington, DC. [Poster]
- 2024 Bista, S., **Bhattacharai, S.**, Sharma, S., & Talchabhadel, R. Implications of non-stationary climate dynamics for flood risk projections (H05-13). AGU Fall Meeting, Washington, DC. [Oral]
- 2024 Banjara, P., **Bhattacharai, S.**, Pandey, V. P., & Talchabhadel, R. Assessing heatwave dynamics and vulnerability in Nepal: A geospatial analysis (H34F-04). AGU Fall Meeting, Washington, DC. [Oral]
- 2024 Bista, S., Bhattacharai, Y., **Bhattacharai, S.**, Sharma, S., Talchabhadel, R., & Amini, F. Urban flood inundation prediction under climate change. ASCE-EWRI World Environmental and Water Resources Congress, Milwaukee, WI. [Oral]
- 2024 Talchabhadel, R., **Bhattacharai, S.**, Bista, S., White, L., & Amini, F. Providing undergraduate students with training on evaluating climate change projections within the framework of open-source software infrastructure. ASCE-EWRI World Environmental and Water Resources Congress, Milwaukee, WI. [Oral]
- 2024 Talchabhadel, R., **Bhattacharai, S.**, & Bista, S. Empowering undergraduate students in remote sensing for climate change projection evaluation using open-source software infrastructure. AGU Chapman Conference, Honolulu, HI. [Poster]
- 2024 Talchabhadel, R., **Bhattacharai, S.**, Banjara, P., & Sharma, S. Projecting heatwave consequences in urban centers of the United States under changing climate. AMS 15th Conference on Environment and Health, Baltimore, MD. [Oral]
- 2024 Thapa, B. R., Talchabhadel, R., & **Bhattacharai, S.** Translating multi-hazard risk assessment to compounding and cascading disaster risk management for resilient infrastructure. 2nd GeoMandu International Conference, Kathmandu, Nepal, November. [Invited]

Workshops Taught and Outreach

- 2025 Lead Instructor, Smartphones4water Workshop: GeoClimate Intelligence Platform for crowdsourced and accessible environmental data analysis.
- 2025 Co-Instructor, AUC Data Science Initiative Faculty Workshop, “Harnessing GeoClimate Intelligence for Research Innovation” (28 registered faculty).
- 2025 Lead Instructor, Python Training Workshop, Jackson State University: hydroclimate and geospatial applications.
- 2025 Program Manager and Facilitator, Mastercard × AUC Data Science Initiative Regional Pitch Stop, Jackson State University.

Professional Service

- Peer review *Communications Earth & Environment* (2), *Earth and Space Science* (1), *Urban Climate* (1).
- Convening Student Convener, AGU Fall Meeting 2024 session NH13B, Coastal Storm Risk Modeling and Analysis via Cloud Computing.
- Memberships American Geophysical Union; European Geosciences Union; American Society of Civil Engineers; Nepal Engineering Council.

Technical Skills

- Modeling MRRpy (developer); GSSHA (WMS); ADCIRC (mesh generation in SMS), coupled to GSSHA; 1D and 2D finite-volume shallow-water solvers (implemented, dam-break problems); HEC-HMS and HEC-RAS (basic).
- Machine learning Tree ensembles (XGBoost, LightGBM, CatBoost), multilayer perceptrons, LSTM, and transformers (scikit-learn, PyTorch); satellite foundation-model embeddings (AlphaEarth); SHAP interpretation; spatial cross-validation.
- Programming Python (NumPy, SciPy, pandas, xarray, rioarray, Dask, Zarr, GeoPandas, rasterio); Numba and CuPy (GPU); Google Earth Engine (Python and JavaScript); Streamlit; working knowledge of HTML, CSS, and JavaScript.
- Software practice Git and GitHub; pytest; GitHub Actions continuous integration; PyPI packaging; MkDocs and Read the Docs; QGIS plugin development; Linux and HPC batch processing.
- Data *Satellite*: GRACE/GRACE-FO, SMAP, IMERG, MODIS, Landsat. *Reanalysis and climate*: ERA5, GLDAS, DAYMET, NEX-GDDP-CMIP6. *Land surface*: SoilGrids, HiHydroSoil, FABDEM, ESA WorldCover. *In situ*: ISMN, SCAN/SNOTEL, USCRN, NEON, USGS streamflow and wells, NOAA tide gauges, DHM and S4W stations (Nepal).
- Statistics Extreme-value and frequency analysis; copulas; quantile mapping and bias correction; homogeneity tests (Pettitt, SNHT, Buishand).

References

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